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NYC 311 Service Requests: Response Times, Complaint Patterns, and Borough-Level Performance

This project analyzes 12 months of NYC 311 service request data, the city's non-emergency complaint system covering issues like noise, illegal parking, heat outages, and street conditions. The analysis covers November 1, 2020, through October 31, 2021. The objective was to evaluate complaint volume, borough-level response times, agency workload distribution, and seasonal trends using SQL in BigQuery, with Microsoft Excel used to build pivot tables and a chart from the final query result.

The dataset contains 2,718,173 complaints across 212 distinct complaint types, fielded by 16 city agencies.

Dataset

The dataset used for this project was: bigquery-public-data.new_york_311.311_service_requests

The dataset contains record-level 311 complaint data, including timestamps for when complaints were created and closed, complaint type, descriptor, borough, responding agency, and resolution status. The public dataset stops updating in November 2021, so this analysis uses the most recent complete 12-month window available.

The date range analyzed was November 1, 2020 through October 31, 2021.

Key Findings

Complaint Volume

There were 2,718,173 total complaints over 12 months, averaging roughly 7,400 complaints per day. These complaints covered 212 distinct complaint types and were handled by 16 city agencies. Noise complaints alone account for roughly 27% of the total 311 volume when grouped.

Top Complaint Types

Noise - Residential led all categories with 372,598 complaints, accounting for 13.71% of total complaints. Illegal Parking ranked second with 295,439 complaints, or 10.87% of the total. HEAT/HOT WATER ranked fourth with 185,927 complaints, or 6.84%, driven mainly by winter housing-code complaints.

The top 15 complaint types account for roughly 65% of all 311 volume.

Borough Performance

Brooklyn led in raw volume with 780,587 complaints, making up 28.72% of all complaints. However, the Bronx generated the most complaints per capita at roughly 409 complaints per 1,000 residents, which was well above Manhattan at about 330 and Staten Island at about 248. Queens had the fastest average response time at 5.0 days. Staten Island was the slowest at 7.3 days, which was 46% longer than Queens. Manhattan's median response time was 1 hour, while its average response time was 124 hours. This shows that a small subset of long-running cases pulled the mean upward. The mean and median tell different stories, and both matter.

Agency Workload

NYPD ranked as the top-volume agency in every borough, largely driven by noise complaints, illegal parking, and blocked driveways rather than traditional law enforcement. HPD ranked

second in the Bronx, Brooklyn, and Manhattan, but did not appear in Staten Island's top three. This is consistent with Staten Island's lower share of multi-family housing.

Manhattan was the only borough where DEP, the Department of Environmental Protection, appeared in the top three. Water-system and sewer complaints in the urban core drove this.

Seasonal Patterns

Noise and Housing complaints follow inverse seasonal cycles. Housing peaks in December at 53,337 complaints and bottoms out in August at 11,408, a 76% drop. Noise climbs from a winter low of 33,852 in February to a summer peak of 89,991 in June.

The two categories effectively trade off as the city's largest complaint driver across the year. Housing dominates November through February, while Noise dominates April through October. This pattern has direct staffing implications: NYPD demand scales up in summer, while HPD demand scales up in winter.

SQL Techniques Used

This project used several SQL techniques, including aggregations with COUNT, SUM, and AVG, GROUP BY with conditional filtering, Common Table Expressions, TIMESTAMP_DIFF for response-time calculations, APPROX_QUANTILES to calculate medians alongside means, RANK() window functions with PARTITION BY for borough-level rankings, QUALIFY to filter window-function output, CASE statements for thematic categorization, and FORMAT_DATE for monthly aggregation.

Excel Analysis

The Query 6 result was exported to CSV and loaded into Excel as a structured table named tbl_Trends. Two pivot tables were built from the data.

The first pivot table was a monthly trend pivot, with months by category, used as the source for a pivot chart. The second pivot table was a category share summary showing total complaints and the percent of total by category.

The pivot chart uses focus-and-context formatting. Noise and Housing lines are emphasized in red and blue at 3-point weight, while the four other categories appear in light gray at 1-point weight. The visual shows the inverse seasonal cycle clearly, with the two emphasized lines crossing in spring 2021.

Reproducing the Analysis

To reproduce the analysis, open Google BigQuery and use the dataset:

bigquery-public-data.new_york_311.311_service_requests

Then run the SQL files in order:

01_total_volume.sql

02_top_complaints.sql

03_borough_volume.sql

04_borough_response.sql

05_agency_ranking.sql

06_monthly_trends.sql

After running Query 6, export the result as a CSV file. Then open the file in Excel, load it as a table, and build pivot tables and a pivot chart from the data.

Tools Used

The tools used for this project were Google BigQuery, SQL, and Microsoft Excel, including tables, pivot tables, and a pivot chart.

Query Walkthrough

Query 1: Total Complaint Volume

The purpose of the first query was to establish the size of the dataset before going deeper.

SQL Query:

```sql
-- Query 1: Total Complaint Volume
-- Purpose: Establish total complaint volume for the analysis period.
-- Date Range: November 1, 2020 – October 31, 2021

SELECT
  COUNT(*) AS total_complaints,
  COUNT(DISTINCT complaint_type) AS unique_complaint_types,
  COUNT(DISTINCT agency) AS responding_agencies
FROM
  `bigquery-public-data.new_york_311.311_service_requests`
WHERE
  created_date BETWEEN '2020-11-01' AND '2021-10-31';
```

Result:

Total complaints: 2,718,173

Unique complaint types: 212

Responding agencies: 16

Interpretation:

Between November 1, 2020, and October 31, 2021, NYC received 2,718,173 complaints through its 311 service request system. These complaints spanned 212 distinct complaint types and were handled by 16 city agencies, ranging from NYPD and HPD to DSNY and DOT. This averages roughly 7,400 complaints per day.

The diversity of complaint types and agencies sets up the rest of the analysis. 311 is not a single uniform service operation, but a coordinated entry point into a large ecosystem of city services, each with its own response patterns.

Query 2: Top Complaint Types

The purpose of the second query was to identify which complaint categories drive the most volume.

SQL Query:

```sql
-- Query 2: Top 15 Complaint Types
-- Purpose: Identify the most common 311 complaint categories.
-- Date Range: November 1, 2020 – October 31, 2021

SELECT
  complaint_type,
  COUNT(*) AS complaint_count,
  ROUND(COUNT(*) * 100.0 / 2718173, 2) AS percent_of_total
FROM
  `bigquery-public-data.new_york_311.311_service_requests`
WHERE
  created_date BETWEEN '2020-11-01' AND '2021-10-31'
GROUP BY
  complaint_type
ORDER BY
  complaint_count DESC
LIMIT 15;
```

Top 5 Results:

Noise - Residential: 372,598 complaints, or 13.71%

Illegal Parking: 295,439 complaints, or 10.87%

Noise - Street/Sidewalk: 188,187 complaints, or 6.92%

HEAT/HOT WATER: 185,927 complaints, or 6.84%

Blocked Driveway: 148,168 complaints, or 5.45%

Interpretation:

Noise complaints dominate the dataset. Residential noise alone accounts for nearly 14% of all 311 calls. When all noise-related categories are combined, including residential, street, vehicle,

commercial, and uncategorized noise complaints, they represent roughly 27% of total volume. That equals more than 742,000 complaints in a single year.

The remainder of the top 15 covers core quality-of-life and infrastructure concerns, including illegal parking, heat and hot water issues, blocked driveways, and street conditions. The top 15 categories account for roughly 65% of all 311 volume, with the remaining 197 complaint types each contributing less than 2%.

The strong representation of noise complaints during this window likely reflects pandemic-era conditions, where increased time at home heightened residents' sensitivity to neighborhood noise. This pattern is revisited in Query 6, which examines monthly trends.

Query 3: Complaints by Borough

The purpose of the third query was to identify which boroughs generate the most complaint volume and add population context.

SQL Query:

```sql
-- Query 3: Complaints by Borough
-- Purpose: Identify which NYC boroughs generate the most 311 complaint volume.
-- Date Range: November 1, 2020 – October 31, 2021

SELECT
  borough,
  COUNT(*) AS complaint_count,
  ROUND(COUNT(*) * 100.0 / 2718173, 2) AS percent_of_total,
  COUNT(DISTINCT complaint_type) AS unique_complaint_types
FROM
  `bigquery-public-data.new_york_311.311_service_requests`
WHERE
  created_date BETWEEN '2020-11-01' AND '2021-10-31'
GROUP BY
  borough
ORDER BY
  complaint_count DESC;
```

Result:

Brooklyn: 780,587 complaints, or 28.72%

Queens: 633,521 complaints, or 23.31%

Bronx: 601,897 complaints, or 22.14%

Manhattan: 558,126 complaints, or 20.53%

Staten Island: 122,800 complaints, or 4.52%

Interpretation:

Brooklyn leads in raw complaint volume, but the rankings shift significantly once population is accounted for. On a per-capita basis, the Bronx files roughly 409 complaints per 1,000 residents,

the highest rate in the city. That is about 55% more than Staten Island per capita and over 65% more than Queens.

This raises a real analytical question: are these complaints reflecting greater service-need density, more responsive civic engagement, or differences in how residents interact with city government?

Approximately 0.79% of complaints have a blank or “Unspecified” borough field. These records were excluded from per-capita interpretations but retained in totals.

Query 4: Average Response Time by Borough

The purpose of the fourth query was to calculate how long the city takes to resolve complaints, broken down by borough. This query reports both mean and median because response-time distributions are typically skewed by long-running cases.

This query introduces a Common Table Expression, or CTE, which is a named subquery used to filter and prepare data before final aggregation.

SQL Query:

```sql
-- Query 4: Average Response Time by Borough
-- Purpose: Calculate how long it takes the city to resolve 311 complaints,
--          broken down by borough. Uses a CTE to filter to closed complaints
--          and compute response time in hours before aggregating.
-- Date Range: November 1, 2020 – October 31, 2021

WITH closed_complaints AS (
  SELECT
    borough,
    TIMESTAMP_DIFF(closed_date, created_date, HOUR) AS response_hours
  FROM
    `bigquery-public-data.new_york_311.311_service_requests`
  WHERE
    created_date BETWEEN '2020-11-01' AND '2021-10-31'
    AND closed_date IS NOT NULL
    AND closed_date > created_date
    AND borough IS NOT NULL
    AND borough != ''
    AND borough != 'Unspecified'
)

SELECT
  borough,
  COUNT(*) AS resolved_complaints,
  ROUND(AVG(response_hours), 1) AS avg_response_hours,
  ROUND(AVG(response_hours) / 24, 1) AS avg_response_days,
  APPROX_QUANTILES(response_hours, 2)[OFFSET(1)] AS median_response_hours
FROM
  closed_complaints
GROUP BY
  borough
ORDER BY
  avg_response_hours ASC;
```

Result:

Queens: 5.0 days average, 2 hours median

Brooklyn: 5.1 days average, 3 hours median

Manhattan: 5.2 days average, 1 hour median

Bronx: 6.4 days average, 4 hours median

Staten Island: 7.3 days average, 22 hours median

Interpretation:

Average response times range from 5.0 days in Queens to 7.3 days in Staten Island. The Bronx and Staten Island stand out as the slowest boroughs, with response times 27% and 46% longer than the fastest borough. Combined with the per-capita finding from Query 3, this raises a service-equity question worth flagging: the borough generating the most service requests per capita, the Bronx, also waits longer for resolution.

The mean-median gap reveals a more nuanced picture. In Manhattan, the median complaint is resolved within one hour, but the average is 124 hours. A small subset of long-running cases pulls the mean upward, while the typical resolution is nearly immediate.

Staten Island's median is 22 hours, meaning the typical complaint takes nearly a full day to resolve before any tail effect is considered. Reporting both metrics matters because relying on averages alone would obscure this pattern. The mean tells us about the worst-case drag, while the median tells us about the typical resident experience.

Query 5: Agency Performance Ranking

The purpose of the fifth query was to rank the top three agencies in each borough by complaint volume.

This query uses a window function, RANK() OVER PARTITION BY, to rank agencies within each borough independently. Window functions compute aggregates without collapsing the underlying rows, which is exactly what borough-specific ranking requires. The QUALIFY clause filters the window output to keep only the top three agencies per borough.

SQL Query:

```sql
-- Query 5: Agency Performance with Borough-Level Ranking
-- Purpose: Rank each agency's complaint volume within each borough using a
--          window function. Reveals which agencies dominate in different
--          parts of the city.
-- Date Range: November 1, 2020 – October 31, 2021

WITH agency_borough_volume AS (
  SELECT
    borough,
    agency,
    agency_name,
    COUNT(*) AS complaint_count
  FROM
    `bigquery-public-data.new_york_311.311_service_requests`
  WHERE
    created_date BETWEEN '2020-11-01' AND '2021-10-31'
    AND borough IS NOT NULL
    AND borough != ''
    AND borough != 'Unspecified'
  GROUP BY
    borough, agency, agency_name
)

SELECT
  borough,
  agency,
  agency_name,
  complaint_count,
  RANK() OVER (PARTITION BY borough ORDER BY complaint_count DESC) AS rank_in_borough
FROM
  agency_borough_volume
QUALIFY
  rank_in_borough <= 3
ORDER BY
  borough, rank_in_borough;
```

Result Highlights:

NYPD ranked first in every borough. Brooklyn NYPD volume alone was 358,408 complaints. HPD ranked second in the Bronx, Brooklyn, and Manhattan. DEP appeared in the top three only in Manhattan. Staten Island's top three agencies were NYPD, DOT, and DSNY.

Interpretation:

NYPD ranks first in every borough, often by a wide margin. This reframes how we think about the NYPD's workload. A substantial portion of its 311 volume comes from noise complaints, illegal parking, blocked driveways, and similar quality-of-life issues, rather than traditional law enforcement matters. NYPD effectively serves as the city's primary front line for non-emergency neighborhood enforcement.

HPD ranks second in three boroughs, reflecting the high volume of heat, hot water, and housing-code complaints in dense rental markets. HPD does not appear in Staten Island's top three, consistent with that borough's lower share of multi-family housing.

Manhattan is the only borough where DEP appears in the top three, driven by water-system and sewer complaints. Staten Island's profile, with NYPD, DOT, and DSNY as the top agencies, reads more like a suburban service mix.

The Bronx absorbs the highest absolute volume of HPD complaints, with 162,779 complaints, despite ranking third in population. Combined with the per-capita complaint volume from Query 3 and the slower response times from Query 4, this points to a borough where housing-related service demand is highest and resolution is slowest.

Query 6: Monthly Trends with Category Bucketing

The purpose of the sixth query was to track how complaint volume changed month-over-month across six high-level categories. Because the dataset contains 212 distinct complaint types, direct analysis at the type level is unwieldy. The CASE statement buckets complaint types into themes that operations teams can plan around.

SQL Query:

```sql
-- Query 6: Monthly Complaint Trends with Category Bucketing
-- Purpose: Track how complaint volume changed month-over-month, broken into
--          high-level thematic categories (Noise, Housing, Sanitation, etc.)
--          using a CASE statement.
-- Date Range: November 1, 2020 – October 31, 2021

SELECT
  FORMAT_DATE('%Y-%m', DATE(created_date)) AS year_month,
  CASE
    WHEN complaint_type LIKE 'Noise%' THEN 'Noise'
    WHEN complaint_type IN ('HEAT/HOT WATER', 'UNSANITARY CONDITION', 'PLUMBING', 'PAINT/PLASTER', 'GENERAL CONSTRUCTION', 'ELECTRIC') THEN 'Housing'
    WHEN complaint_type IN ('Illegal Parking', 'Blocked Driveway', 'Abandoned Vehicle', 'Derelict Vehicles', 'Street Condition', 'Street Light Condition') THEN 'Streets & Vehicles'
    WHEN complaint_type IN ('Missed Collection (All Materials)', 'Dirty Conditions', 'Sanitation Condition', 'Overflowing Litter Baskets') THEN 'Sanitation'
    WHEN complaint_type IN ('Water System', 'Sewer') THEN 'Water/Sewer'
    ELSE 'Other'
  END AS complaint_category,
  COUNT(*) AS complaint_count
FROM
  `bigquery-public-data.new_york_311.311_service_requests`
WHERE
  created_date BETWEEN '2020-11-01' AND '2021-10-31'
GROUP BY
  year_month, complaint_category
ORDER BY
  year_month, complaint_count DESC;
```

Headline Finding:

Noise and Housing complaints follow inverse seasonal cycles. Housing peaks in December at 53,337 complaints and bottoms out in August at 11,408, a 76% drop. Noise climbs from 33,852 in February to 89,991 in June. The two categories alternate as the city's largest complaint driver across the year, with Housing dominating November through February and Noise dominating April through October.

This pattern has clear operational implications. NYPD demand, which handles noise, scales up in summer. HPD demand, which handles heat and housing-code complaints, scales up in winter. Capacity planning for the two agencies needs to run on opposite schedules.

Streets & Vehicles complaints stay comparatively stable across the year, ranging from roughly 44,000 to 64,000 per month. Sanitation, Water/Sewer, and Other show smaller seasonal variation but contribute meaningful baseline volume.

A small data quality observation worth flagging is that October 2021 shows an unusually high count in the "Other" bucket, with 92,445 complaints. This suggests an emerging complaint type or category not captured by the original CASE mapping. In an operational analytics environment, this kind of spike is a signal that the complaint taxonomy should be reviewed periodically so new issue types get categorized rather than swept into a generic "Other" bin.

Conclusion

This analysis surfaced three findings that connect across the six queries.

First, quality-of-life issues drive 311 volume. Noise complaints alone account for roughly 27% of all 311 calls, with illegal parking, blocked driveways, and heat/hot water rounding out the top of the distribution. These complaints route disproportionately through the NYPD, which ranks as the top-volume agency in every borough. NYPD's 311 workload is dominated by non-policing matters.

Second, borough-level service disparities exist. The Bronx files the most complaints per capita and waits longer than average for resolution. Staten Island's typical response time, with a 22-hour median, is structurally different from Manhattan's 1-hour median. These patterns are not visible if you only look at city-wide averages. They require borough-level breakdowns plus mean-and-median reporting to surface.

Third, seasonal patterns have staffing implications. The Noise-Housing inverse cycle is the most actionable pattern in the data. The 76% drop in Housing from peak to trough, combined with the matching summer surge in Noise, shows that two of the city's largest service demands run on

opposite schedules. A city operations manager could use this to plan agency-level capacity month by month.

Compared to my first portfolio project, this one introduced four SQL techniques I had not used before: Common Table Expressions, window functions, CASE statements for derived categorization, and APPROX_QUANTILES for median calculation. Combining these with Excel pivot analysis demonstrated the full analyst workflow: extracting and shaping data with SQL, then summarizing and visualizing it for non-technical audiences.

The most useful lesson was the analytical importance of reporting both mean and median when distributions are skewed. A single number can obscure the typical experience of users in ways that distort decisions made on top of the data.

Works Cited

Google Cloud. “NYC 311 Service Requests.” BigQuery Public Datasets, Google Cloud, dataset: bigquery-public-data.new_york_311.311_service_requests. Accessed 24 Apr. 2026.